



MISSOURI

S&T

MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Founded 1870 | Rolla, Missouri

Section 2.3

Characterizations of Invertible Matrices

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent.

- A is an invertible matrix.
- A is row equivalent to I_n .
- A has n pivot positions.
- The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- The columns of A form a linearly independent set.
- The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent.

- The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
- The columns of A span $\mathbb{R}^n$.
- The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- There is an $n \times n$ matrix C such that $CA = I$.
- There is an $n \times n$ matrix D such that $AD = I$.
- A^T is an invertible matrix.

Fact

Let A and B be square $n \times n$ matrices.
If $AB = I$, then A and B are both invertible,
with $B = A^{-1}$ and $A = B^{-1}$.

Comments

Each statement in the Invertible Matrix Theorem describes a property of every $n \times n$ invertible matrix.

The negation of a statement in the theorem describes a property of every $n \times n$ singular matrix.

Example 1

Determine if each matrix is invertible.

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 4 & 5 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$$

Example 2

If C is 6×6 and the equation $C\mathbf{x} = \mathbf{v}$ is consistent for every $\mathbf{v}$ in $\mathbb{R}^6$, is it possible that for some $\mathbf{v}$, the equation $C\mathbf{x} = \mathbf{v}$ has more than one solution? Why or why not?

Example 3

If A is an $n \times n$ matrix, then the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.

Example 4

Show that if AB is invertible, so is B .

Caution About Homework

For the Section 2.3 homework, assume that all matrices are $n \times n$ unless otherwise specified.

Work this problem assuming A and B are square.

Invertible Linear Transformations

A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible if there exists a function $S: \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that

$$S(T(\mathbf{x})) = \mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

and

$$T(S(\mathbf{x})) = \mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

Theorem

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T . Then, T is invertible if and only if A is an invertible matrix. In that case, $S(\mathbf{x}) = A^{-1}\mathbf{x}$ satisfies the previous conditions.

Example 5

Show that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is invertible and find a formula for T^{-1} .

$$T(x_1, x_2) = (-5x_1 + 9x_2, 4x_1 - 7x_2)$$
